```
# (e) exact values in Q(sqrt5): cycle matrix and coefficients computed exactly from
the integer table.
import numpy as np, io, contextlib, sympy as sp
from sympy.polys.matrices import DomainMatrix
ns={}
with contextlib.redirect_stdout(io.StringIO()):
exec(open('45_table_only_cycle_frame.py').read(),ns)
G,B,res,N=ns['G'],ns['B'],ns['res'],ns['N']; p1=res[0][1]
K=sp.QQ.algebraic_field(sp.sqrt(5)); s5=K.from_sympy(sp.sqrt(5))
GBq=sp.Matrix(G[np.ix_(B,B)].tolist()); GBinv=GBq.inv()                      # exact
rational
Cq=(GBinv*sp.Matrix(G[B,:].tolist())).T                                     # exact
rational coefficients of all 240
print("exact coefficients reproduce table:", (Cq*GBq*Cq.T)==sp.Matrix(G.tolist()))
M=sp.Matrix([[Cq[p1[B[b]],i] for b in range(8)] for i in range(8)])          # column
b = image of basis element b
x=sp.symbols('x'); cp=M.charpoly(x).as_expr(); print("characteristic polynomial
(rational):",sp.factor(cp))
fs=sp.factor_list(cp,extension=sp.sqrt(5))[1]
D=lambda A: DomainMatrix.from_Matrix(A).convert_to(K)
MD=D(M); GBD=D(GBq); CD=D(Cq)
for f,_ in fs:
    co=sp.Poly(f,x).all_coeffs(); F=DomainMatrix.zeros((8,8),K)
    for c in co: F=F*MD+DomainMatrix.eye(8,K)*K.from_sympy(c)
    Kn=F.nullspace().transpose()                                            # 8x4 exact
basis of the sector
    Q=GBD*Kn*(Kn.transpose()*GBD*Kn).inv()*Kn.transpose()*GBD
    V=(CD*Q*CD.transpose())                                                 # V[r,r] =
exact squared sector value
    vals=sorted({K.to_sympy(V[r,r].element) for r in range(N)},key=float)
    print("factor",sp.simplify(f),"\n   exact squared sector values:",
[sp.nsimplify(v) for v in vals],
          "\n   counts:",[sum(1 for r in range(N) if K.to_sympy(V[r,r].element)==v)
for v in vals])
    if len(vals)==2:
        a,b=vals; print("   b/a =",sp.radsimp(b/a)," | equals
phi^2:",sp.simplify(b/a-((1+sp.sqrt(5))/2)**2)==0," | a+b =",sp.simplify(a+b))
```